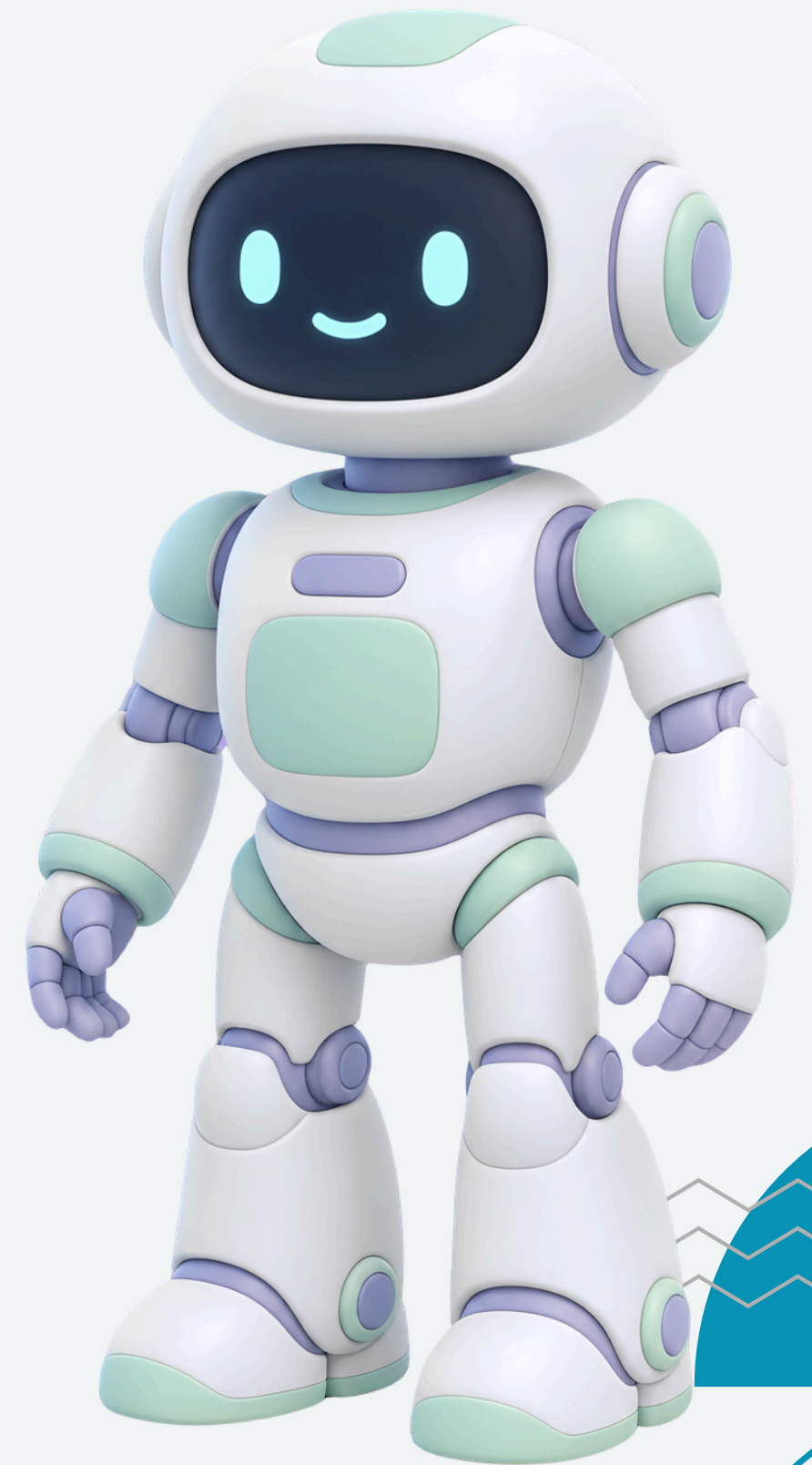


If the hospital is huge... and you don't know
where to go or who to ask



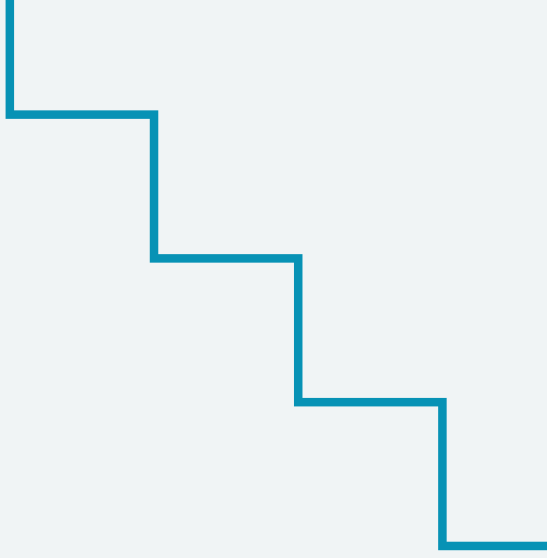
ROMI

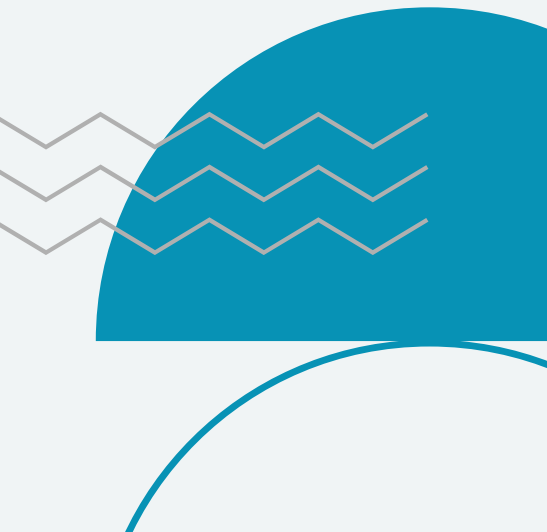
Robotic Medical Intelligence

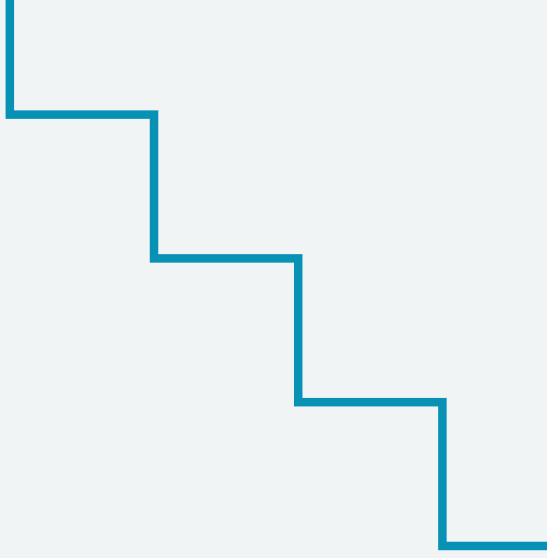




Team Members



- YOUSSEF SALEH
 - HONOOD RAMADAN
 - ROAA AMR
 - MARIA MILAD
 - LAMAR ALAA
- 
- 



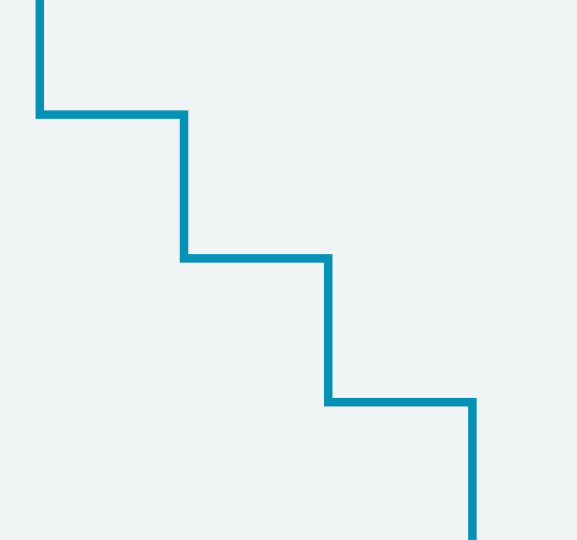
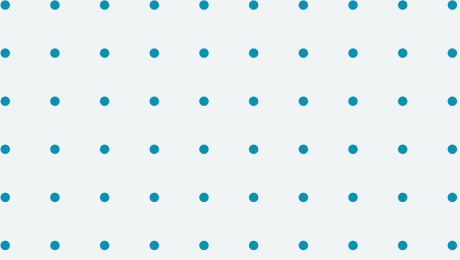
THE PROBLEM

1. Problem Statement:

- Manual reception causes overcrowding, long waits, and administrative burnout.

2. Target Users:

- Clinic Owners & Administrators: Cost reduction & efficiency.
 - Patients: Fast & secure check-ins.
 - Receptionists: Less repetitive work.
- 
- 



RESEARCH & VALIDATION

1. Research Findings :

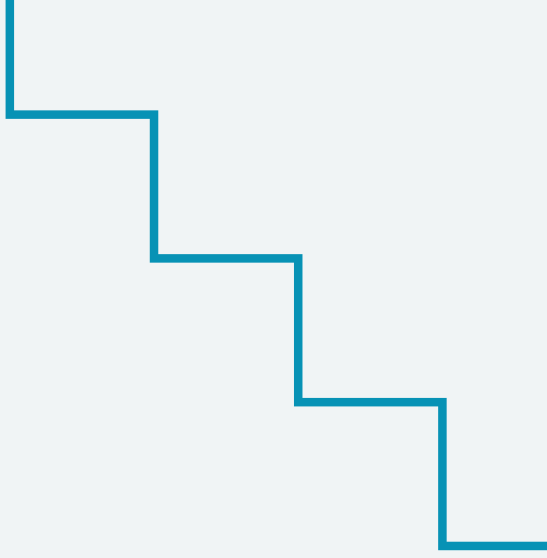
- Overcrowding: Manual check-ins increase patient waiting times.
- Wasted Resources: Receptionists spend 60%+ of shifts on routine verification.

2. References :

- Healthcare reports on patient flow.
 - Interviews and observations from local clinics.
- 
- 



PROPOSED SOLUTION (ROMI)



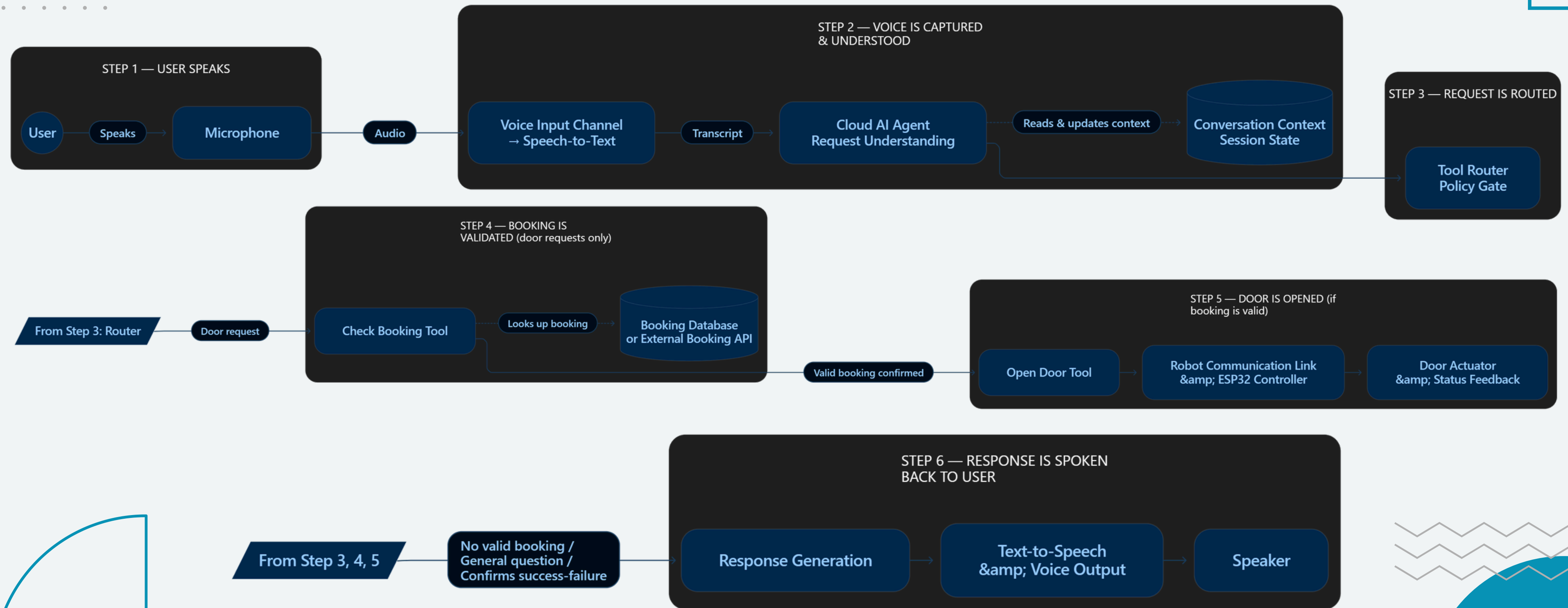
1. What is ROMI?

- An autonomous voice-powered AI receptionist that validates appointments and unlocks doors via IoT

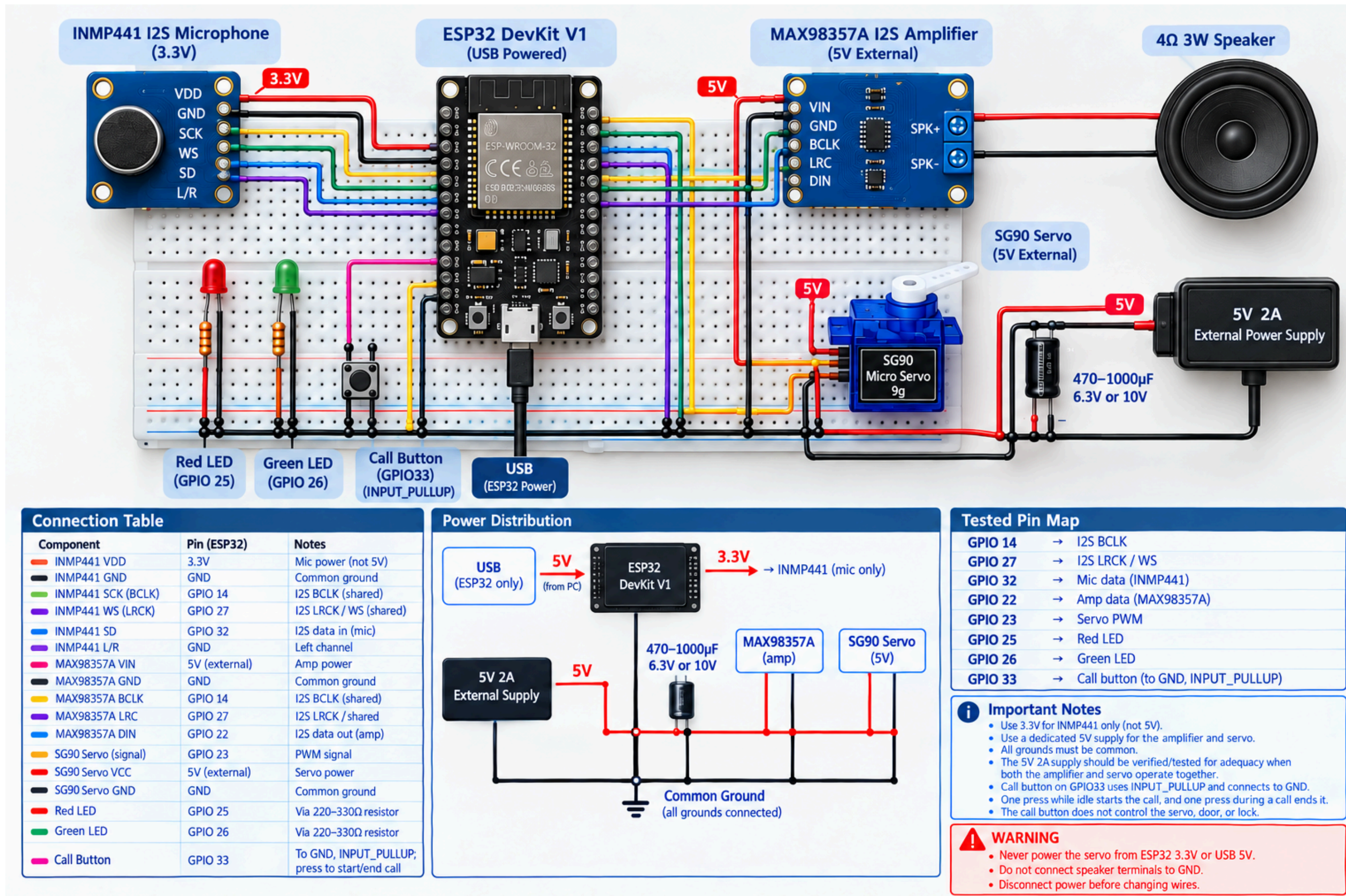
2. Key Features :

- Voice AI: Natural interaction.
 - Live Validation: Instant booking check.
 - Automated IoT Access: ESP32-powered door unlocking.
 - Touchless: Safe, contact-free experience.
- 
- 

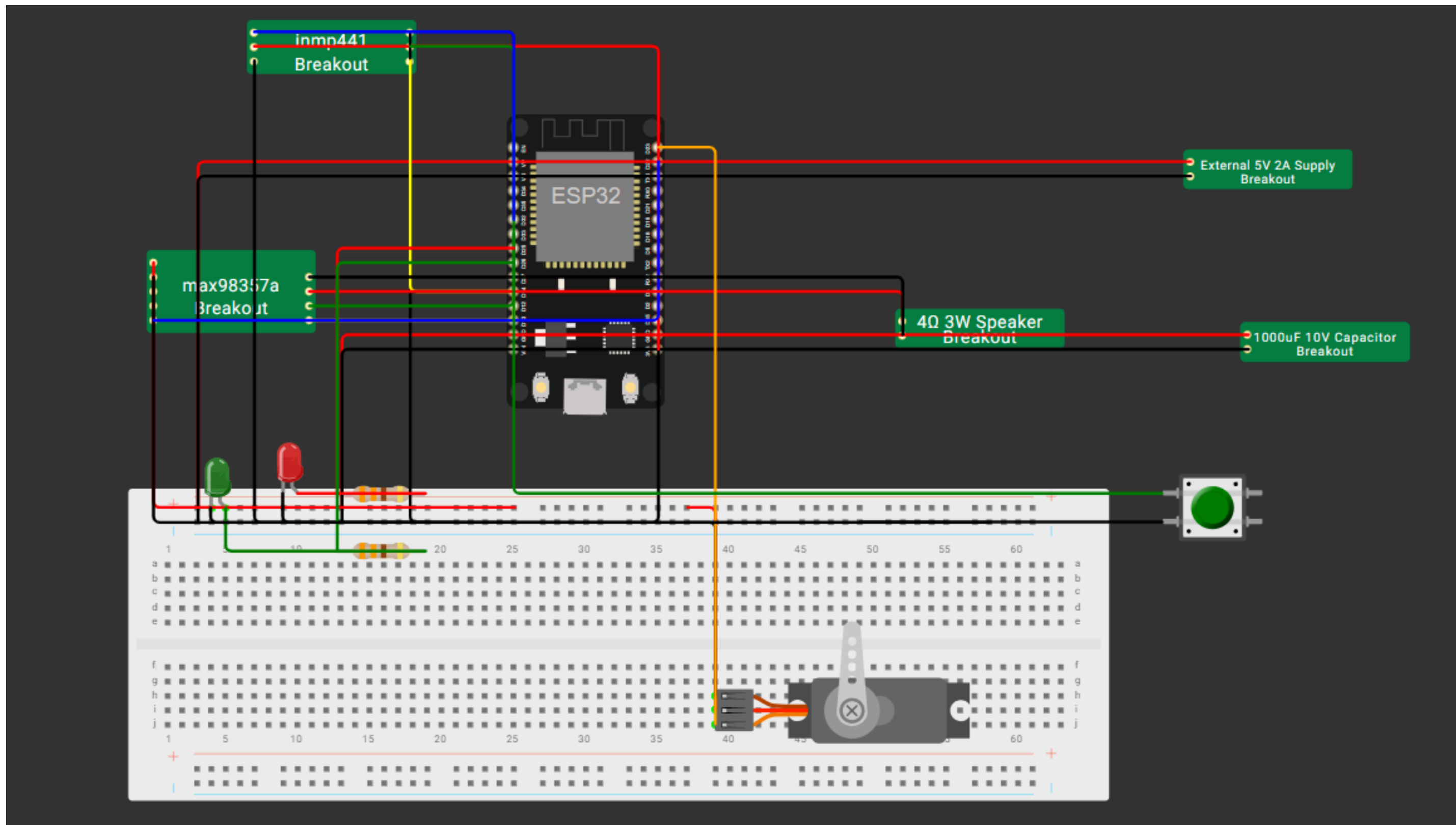
SYSTEM DESIGN



HARDWARE & IOT



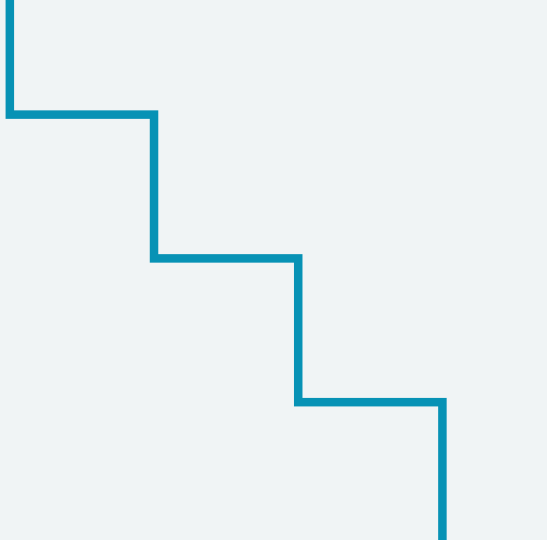
HARDWARE SIMULATION (WOKWI)

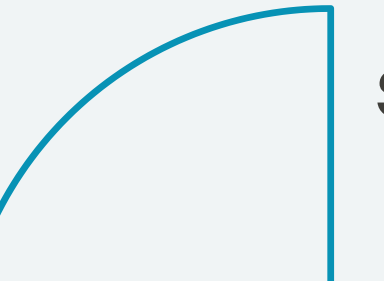
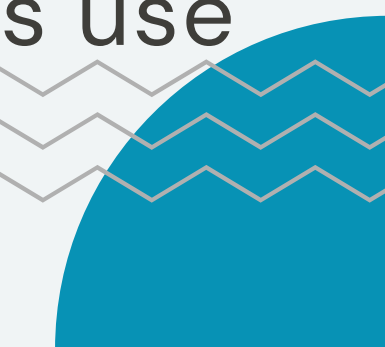


Project: <https://wokwi.com/projects/476639706867674113>



TECHNOLOGY STACK & COMMUNICATION



- Web & Backend: Next.js, React, TypeScript and Tailwind CSS, deployed on Vercel.
 - Database & Authentication: Supabase PostgreSQL, Supabase Auth and Row Level Security.
 - Voice AI: Gemini 3.8 Live (gemini-3.8-live) for real-time Egyptian Arabic speech and function calling.
 - Embedded Controller: ESP32 with C++/Arduino and PlatformIO; I2S microphone/speaker, servo, button and status LEDs.
 - Communication: HTTPS/JSON for backend requests; direct TLS WebSocket for Gemini audio. Device requests use a bearer token; voice sessions use short-lived credentials.
- 
- 

AI TOOLS & API CONTROL

AI Tool	Purpose	API Endpoint
find_available_slots	Find available appointments	/api/ai/availability
create_booking	Create a booking and return its code	/api/ai/create-booking
check_in_booking	Validate a booking code and entry window	/api/ai/check-in
get_booking_entry_status	Refresh current entry eligibility	/api/ai/entry-status
end_conversation	Close the current voice session	Local client action

WHY WE CHOSSE THAT?

1. *ESP32:*

- The main controller for ROMI; manages hardware and provides Wi-Fi connectivity with the web system.

2. *Servo Motor:*

- Controls the hospital door and simulates automatic opening.

3. *Microphone:*

- Receives patients' questions via voice.

4. *Speaker:*

- Responds to patients verbally.

5. *LED:*

- Indicates ROMI's current status.

6. *Resistors:*

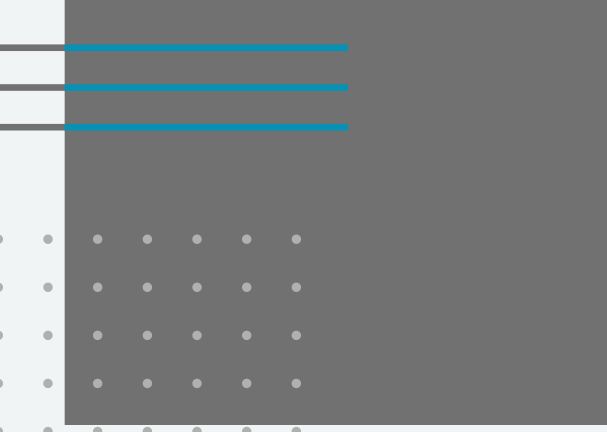
- Protect electronic components and regulate current.

7. *Breadboard & Jumper Wires:*

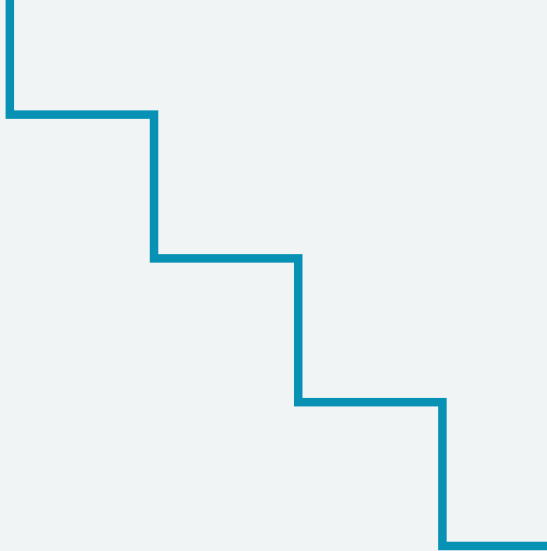
- Connect components and allow easy testing and modification of the circuit.

8. *MAX98357A 12S Amplifier :*

- Digital Audio Output for ESP32 Converting audio streams to clear sound



THE PROBLEMS WE FACED



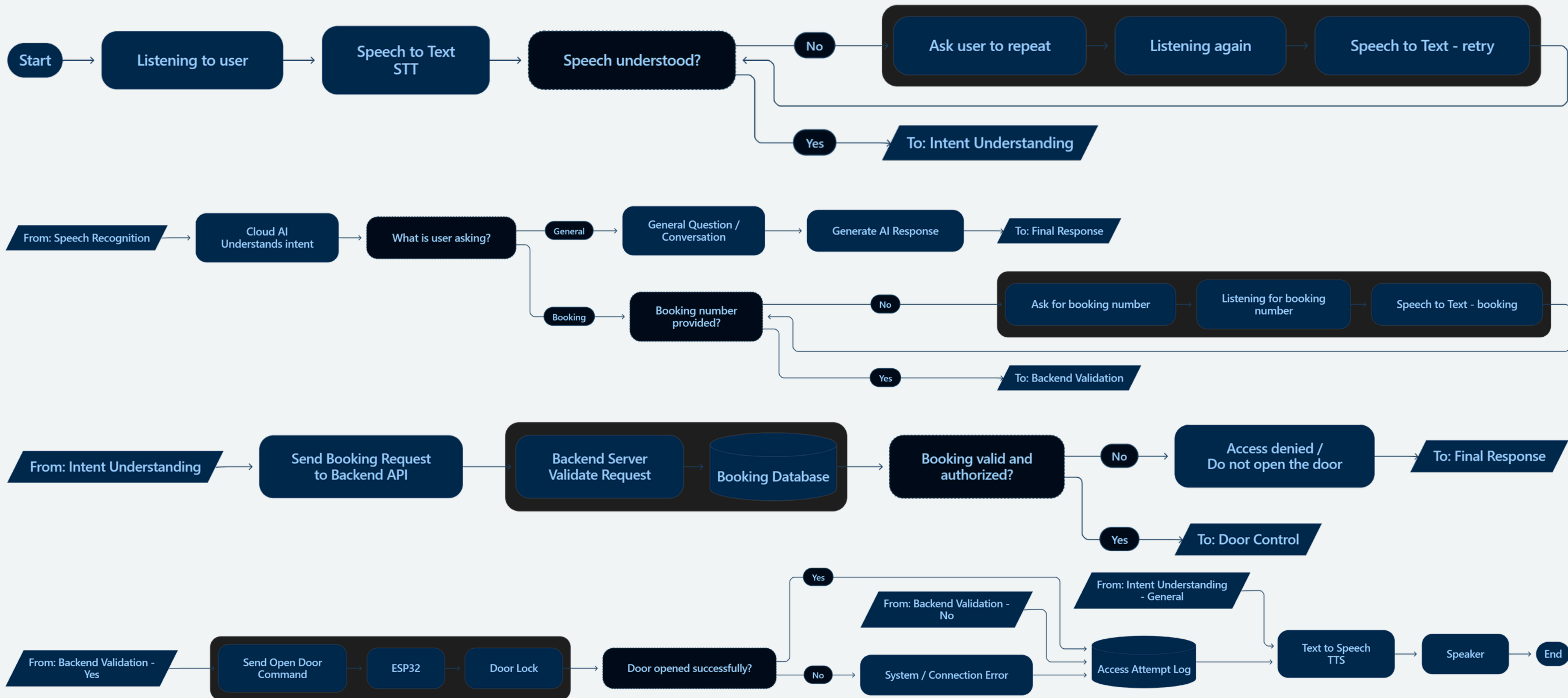
One of our biggest challenges was the hardware, especially:

- Connecting the components correctly.
- Understanding the proper connections and signals.
- Getting all components to work together.
- Finding a practical way to implement our idea.

We solved these issues through experimentation, research, and troubleshooting. This taught us a lot about hardware, circuits, and connections, and showed us that turning an idea into a real project requires constant experimentation and adjustment.

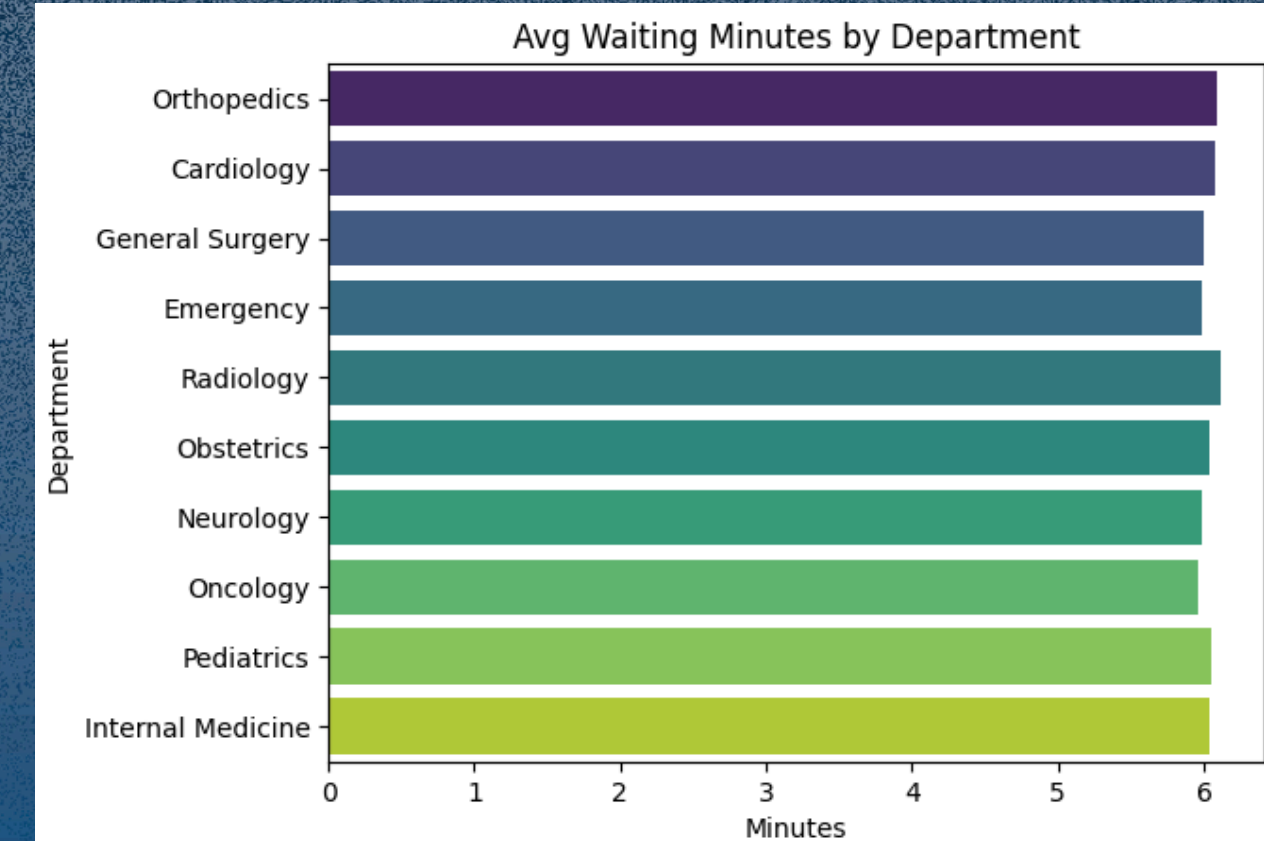
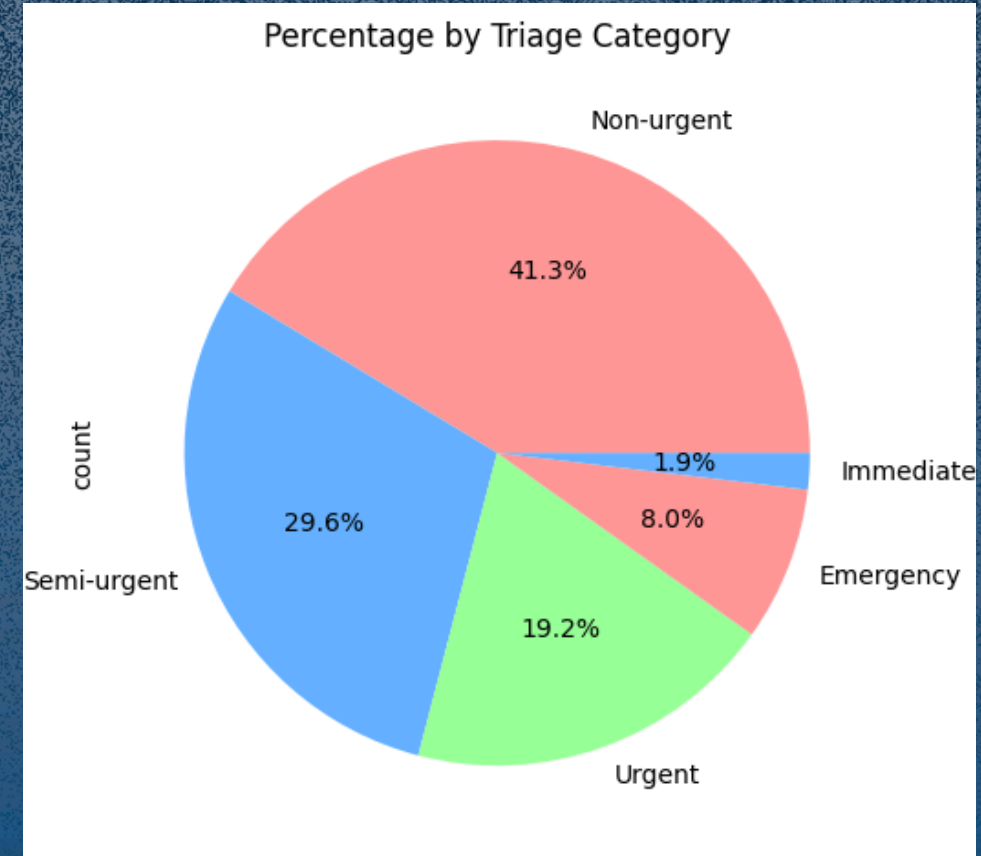
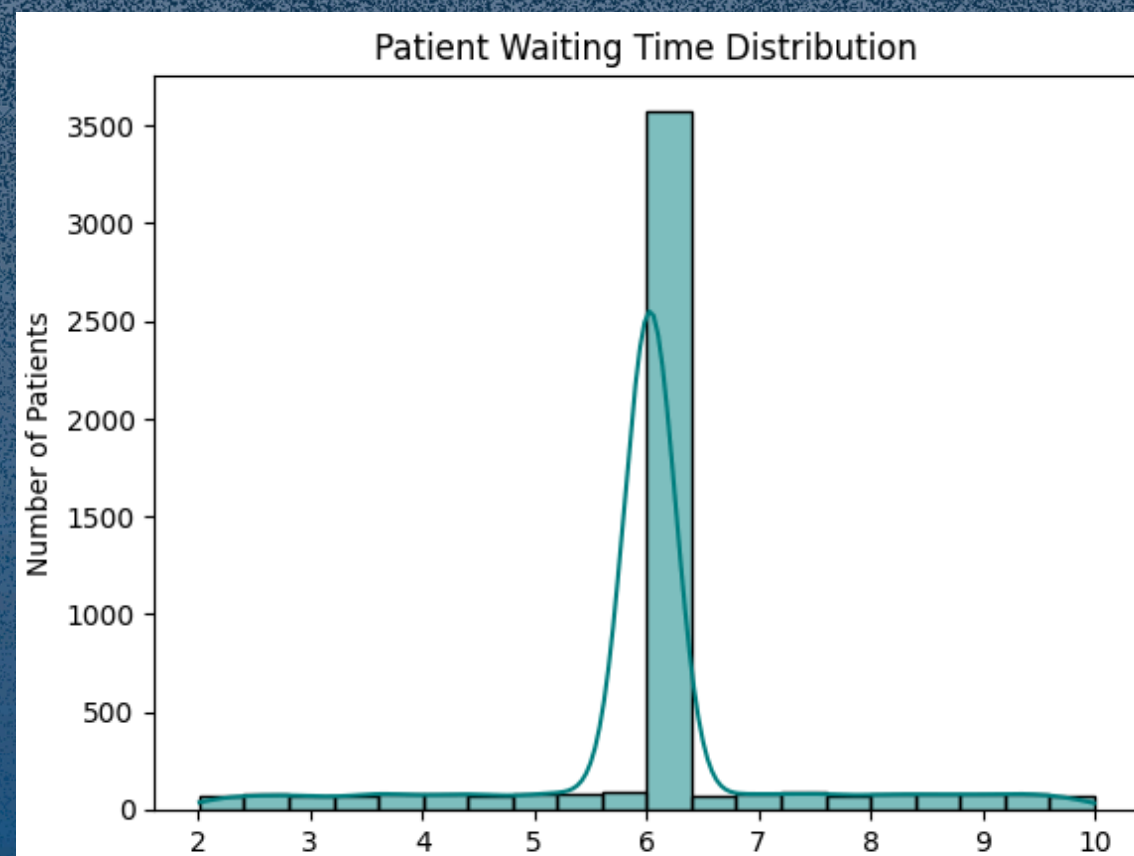


PROGRAMMATIC LOGIC

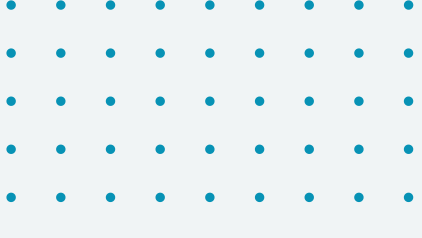


DATA-DRIVEN RESEARCH & PREDICTIVE AI

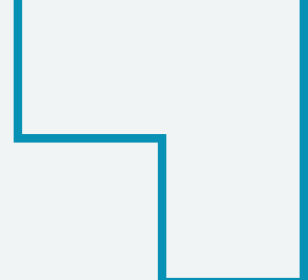
- Dataset Analysis: Analyzed 5,000 hospital records to identify registration wait times as the main bottleneck.
- Predictive ML Model: Trained a RandomForest model to calculate expected patient wait times.
- ROMI Integration: ROMI informs patients of their estimated wait time after



Live Code Evidence: [On Google Colab](#)



AI & DATA USAGE PLAN



1. Data Privacy & Handling :

- Zero Audio Retention: Audio is processed in memory and immediately discarded.
- Structured Extraction: AI extracts only necessary data.

2. AI Decision Engine (Tool Calling) :

- Deterministic Control: Uses Function Calling to trigger actions.
 - Booking Check: Verifies appointments through the database.
 - Door Opening: ESP32 receives a signal only after confirmation.
 - Anti-Hallucination: Access requires verified database confirmation.
- 

DIGITAL IDENTITY

Name:

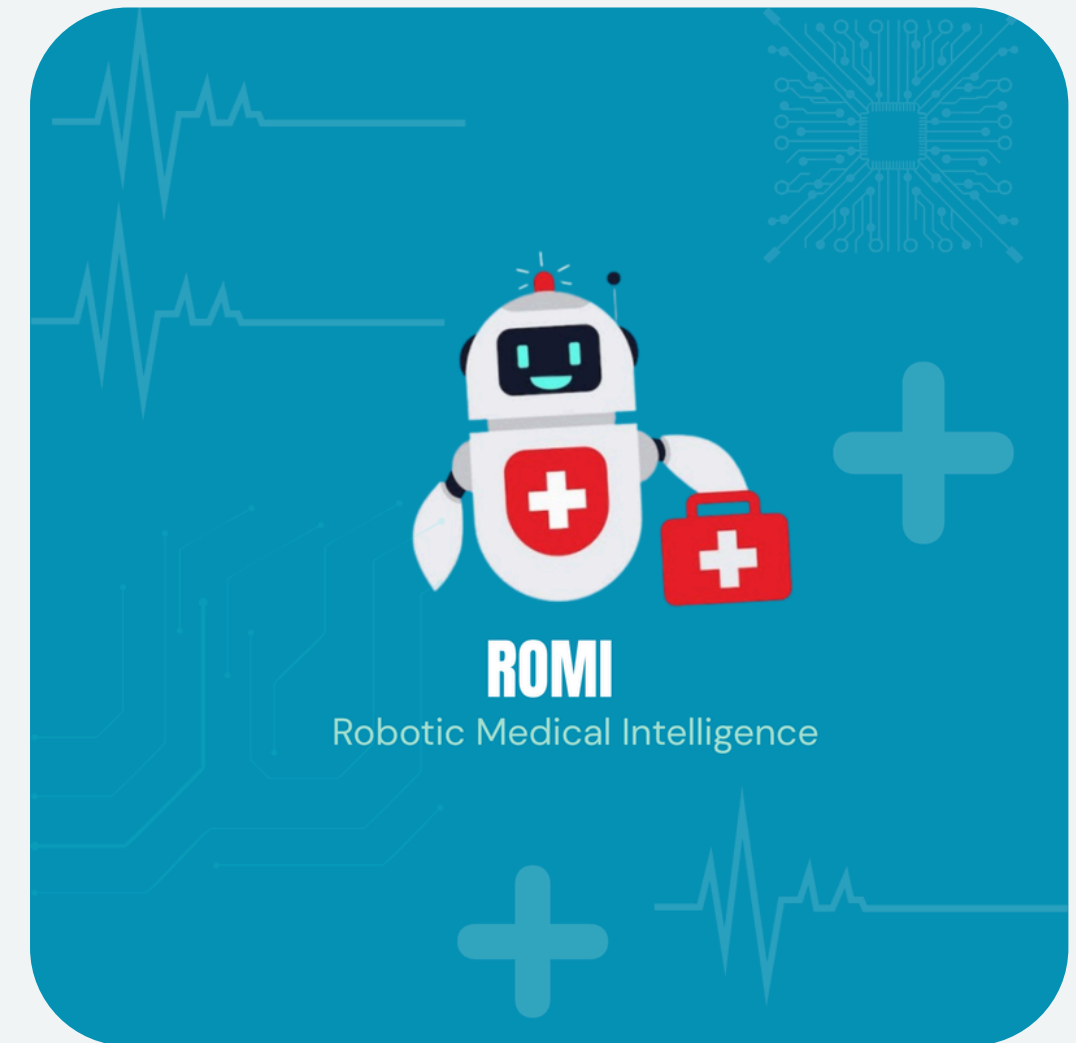
ROMI (Robotic Medical Intelligence).

Identity:

Friendly, reliable, clinical-grade AI assistant.

Colors:

- Teal & Mint: Tech reliability and patient comfort.
- Medical Red: Healthcare identity and urgency.
- Pure White: Hygiene and accessibility.



STYLE GUIDE LINK : <https://canva.link/lwhjqkzwsgdodhk>

TRACK INTEGRATION

- *Digital Arts (UI/UX):*
- Created ROMI's identity and Figma dashboard.
- *Web Development:*
- Built the system architecture and Next.js web app.
- *Data Science:*
- Researched medical technology and analyzed patient-flow data.
- *Cybersecurity:*
- Designed ESP32 circuits and secured the system against:
 1. AI Prompt Injection: AI cannot directly control hardware.
 2. Unauthorized API Access: Only API requests with secret Token are accepted.
 3. Data Exposure: RLS protects patient records.

Live Demo: romi-deci.vercel.app

BUSINESS MODEL

1. Target Customers (B2B):

- Private Clinics & Hospital Outpatient Departments.

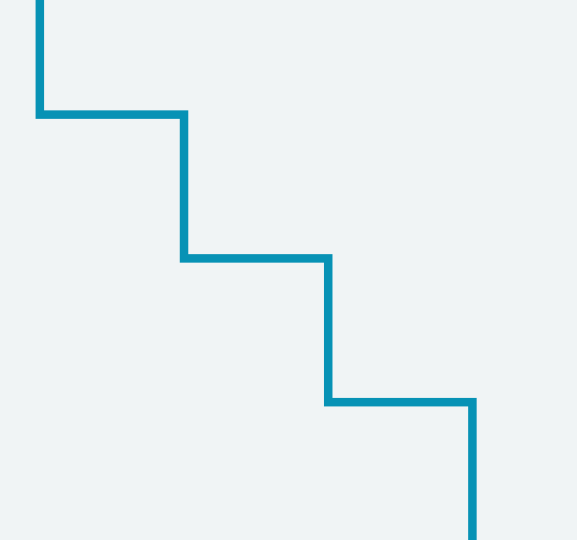
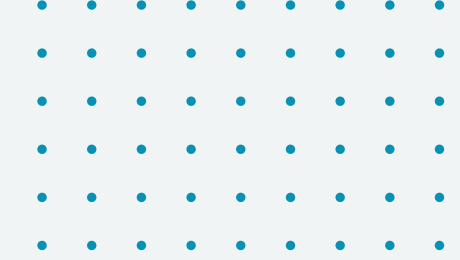
2. Value Proposition (ROI):

- Cost Reduction: Cuts receptionist workload by 60%.
- Efficiency: Eliminates waiting room queues and data errors.
- Brand Image: Tech-forward, premium patient experience.

3. Revenue Streams (Our Profit):

- Hardware Margin: One-time profit from the initial kiosk setup.
- SaaS Subscription: Monthly recurring revenue after LLM/DB costs.
- Maintenance: Annual fees for support and update.



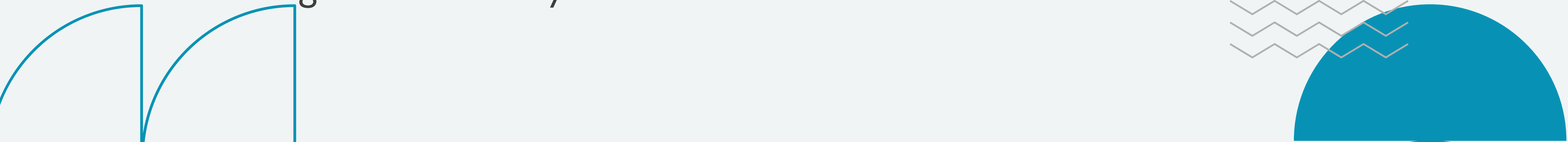


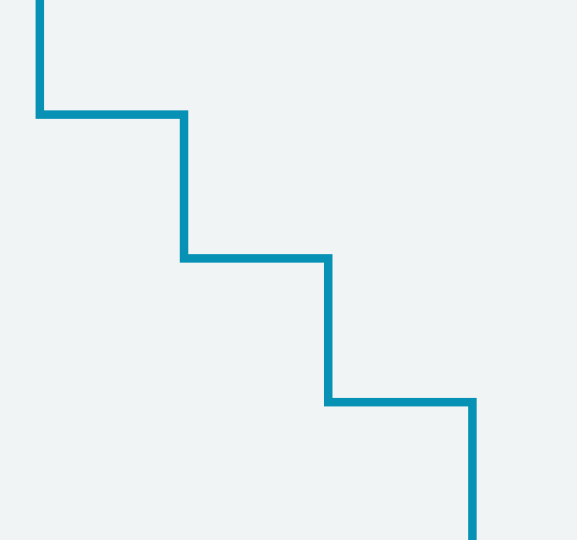
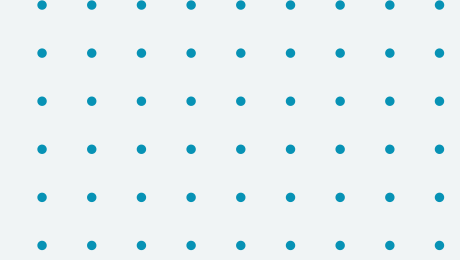
EXTRA POINTS (BONUS)

1. Advanced AI Feature: Multi-Turn Memory & Tool Calling:

- Dynamic Flow: Handles Arabic dialogue and asks follow-up questions when booking details are missing.
- Structured Output: Converts voice requests into precise JSON function calls for secure backend validation

2. Advanced System Feature: Hardware Error Handling & Logging:

- Robust Fallbacks: Instantly detects ESP32 connection drops and alerts the patient via TTS without system crashes.
 - Traceability: Every AI decision and door trigger is securely logged in real-time for clinical-grade security.
- 



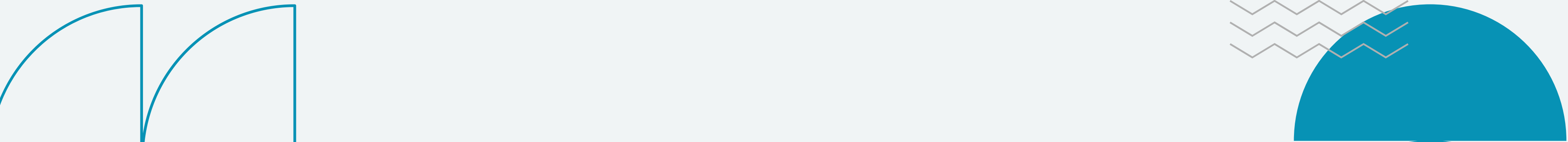
WHY (ROMI) IS THE BEST ?

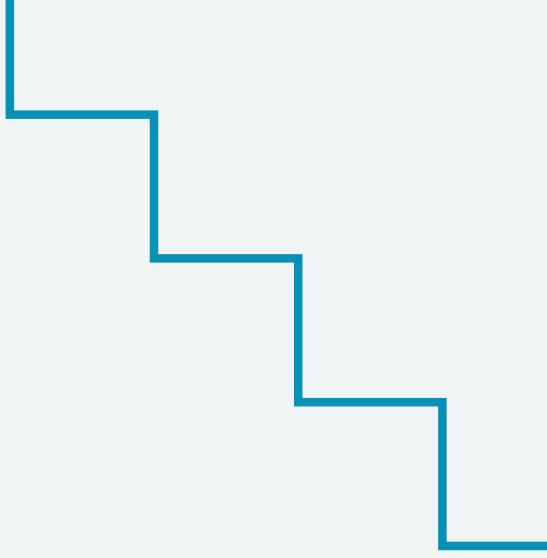
1. ROMI functions as a smart assistant that can:

- Answer frequently asked questions.
- Direct and guide patients.
- Assist with appointments and bookings.

This reduces the workload on reception staff, allowing them to focus on cases that require human intervention and decision-making.

ROMI = Human Support + Smart Technology

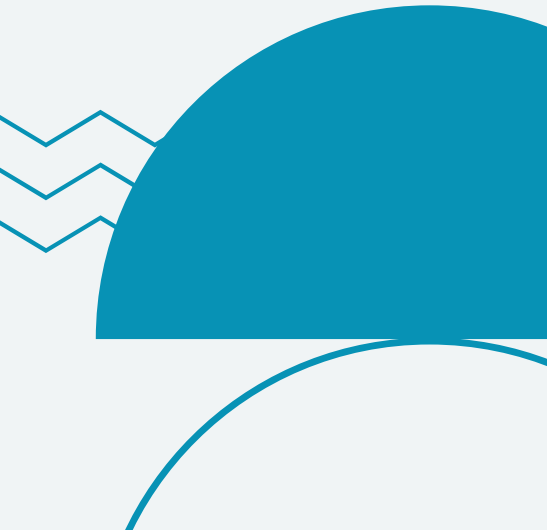




OUR GOAL

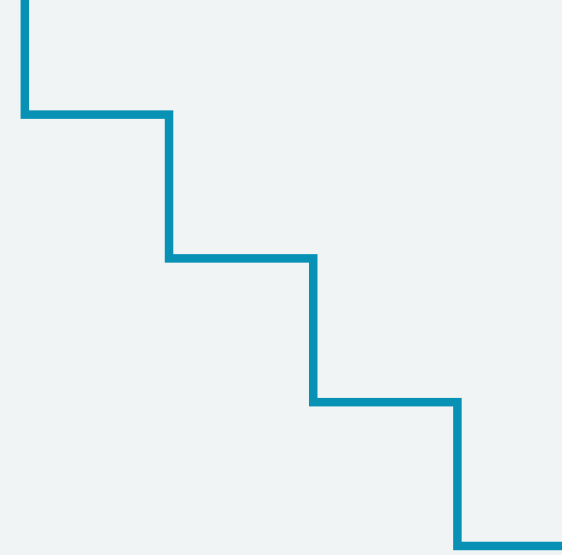
- The goal isn't to replace humans the goal is to use technology for tasks that can be automated, while leaving tasks that require judgment and human interaction to people.

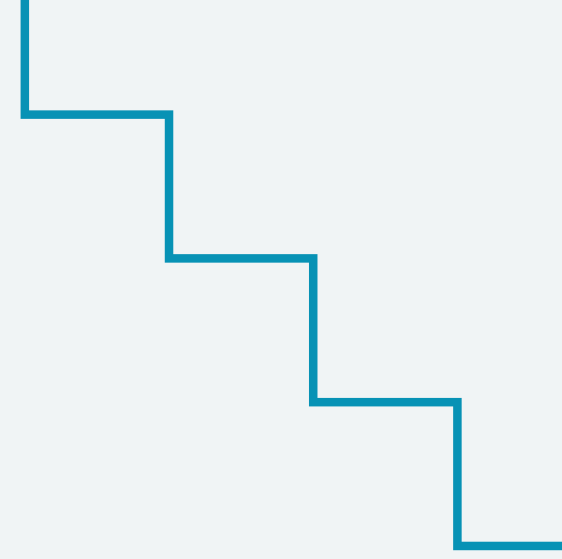
ROMI – Technology that supports people, not replaces them.





Q & A





THANK
YOU

